

Topic Covered:

- Data summarization

1. The number of natural gas drilling sites in Pennsylvania counties with drilling operations is listed below. Find the mean, median, mode and midrange for the data and discuss the differences in these measures of central tendency. Which best describes the center of the data?

9	112	1	1	6	1008	109	6	14	58
20	136	85	3	58	185	6	415	2	43
26	2	2	2	473	53	76	19	41	462
693	1	2	560	4	191	98			

Answer:

Mean: $\frac{\sum x_i}{n} = \frac{4982}{37} \approx 134.65$

Median: For odd n , middle value (19th of 37) = 41

Mode: Most frequent value = 2 (appears 5 times)

Midrange: $\frac{\text{Min} + \text{Max}}{2} = \frac{1 + 1008}{2} = 504.5$

2. Shown are the ten U.S. cities (with population over 100,000) with the fewest number of violent crimes reported in 2010. Find the mean, median, mode, and midrange of the data.

Temecula, CA	74	Irvine, CA	120
Murrieta, CA	99	Cary, NC	120
Surprise, AZ	106	Amherst, NY	121
Round Rock, TX	116	Simi Valley, CA	129
Frisco, TX	119	Norman, OK	132

Answer:

Mean: $\frac{\sum x_i}{n} = \frac{1136}{10} \approx 113.6$.

Median: For even n , average two middle values: $\frac{119 + 120}{2} = 119.5$.

Mode: Most frequent value = 120 (appears 2 times).

Midrange: $\frac{\text{Min} + \text{Max}}{2} = \frac{74 + 132}{2} = 103$.

3. A random sample of bonuses (in millions of dollars) paid by large companies to their executives is shown. Find the mean and modal class for the data.



Class boundaries	Frequency
0.5–3.5	11
3.5–6.5	12
6.5–9.5	4
9.5–12.5	2
12.5–15.5	1

Answer:

$$\text{Mean: } \frac{\sum(f_i \cdot m_i)}{\sum f_i} = \frac{(11 \cdot 2) + (12 \cdot 5) + (4 \cdot 8) + (2 \cdot 11) + (1 \cdot 14)}{30} = \frac{150}{30} = 5 \quad [\text{where } m_i \text{ is class midpoint}]$$

Modal Class: Class with highest frequency (12) = 3.5–6.5

4. Find the weighted mean price of three models of automobiles sold. The number and price of each model sold are shown in this list.

Model	Number	Price
A	8	\$10,000
B	10	12,000
C	12	8,000

Answer:

$$\text{Mean: } \frac{\sum(n_i \cdot p_i)}{\sum n_i} = \frac{(8 \cdot 10,000) + (10 \cdot 12,000) + (12 \cdot 8,000)}{30} = \frac{296,000}{30} \approx 9866.67 \quad [\text{where } n_i \text{ is the number sold}]$$

5. An instructor grades exams, 20%; term paper, 30%; final exam, 50%. A student had grades of 83, 72, and 90, respectively, for exams, term paper, and final exam. Find the student's final average. Use the weighted mean.

Answer:

$$\text{Mean: } \frac{\sum(w_i \cdot x_i)}{\sum w_i} = \frac{(0.20 \cdot 83) + (0.30 \cdot 72) + (0.50 \cdot 90)}{1.00} = \frac{83.2}{1.00} \approx 83.2 \quad [\text{where } w_i \text{ is the weight}]$$

6. If the mean of five values is 64, find the sum of the values.

Answer:

$$\text{Mean} = \frac{\sum x_i}{n} \Rightarrow 64 = \frac{\sum x_i}{5} \Rightarrow \sum x_i = 64 \cdot 5 = 320$$

7. The average weekly unemployment benefits (in dollars) for a random selection of states are listed below. Calculate the range, variance, and standard deviation for the data.

239 214 327 416 321 289 209 356
190 272 252 272 310 276 251

Answer:

$$\text{Range: } R = \text{Max} - \text{Min} = 416 - 190 = 226$$

$$\text{Variance: } s^2 = \frac{\sum(x_i - \bar{x})^2}{n-1} = \frac{\sum\left(x_i - \frac{4194}{15}\right)^2}{15-1} = \frac{\sum(x_i - 279.6)^2}{14} \approx 3577.69$$

$$\text{Standard Deviation: } s = \sqrt{s^2} = \sqrt{3577.6857} \approx 59.8134$$



8. A random sample of 30 states shows the number of low-power FM radio stations for each state. Find the variance and standard deviation for the data.

Class limits	Frequency
1–9	5
10–18	7
19–27	10
28–36	3
37–45	3
46–54	2

Answer:

$$\text{Variance: } s^2 = \frac{n(\sum f_i m_i^2) - (\sum f_i m_i)^2}{n(n-1)} = \frac{30(19,902) - 672^2}{30(30-1)} \approx 167.21$$

$$\text{Standard Deviation: } s = \sqrt{s^2} = \sqrt{167.21} \approx 12.93$$

9. Eighty randomly selected batteries were tested to determine their lifetimes (in hours). The following frequency distribution was obtained. Find the variance and standard deviation for the data. Can it be concluded that the lifetimes of these brands of batteries are consistent?

Class boundaries	Frequency
62.5–73.5	5
73.5–84.5	14
84.5–95.5	18
95.5–106.5	25
106.5–117.5	12
117.5–128.5	6

Answer:

$$\text{Variance: } s^2 = \frac{n(\sum f_i m_i^2) - (\sum f_i m_i)^2}{n(n-1)} = \frac{80(752,621) - 7673^2}{80(80-1)} \approx 211.2$$

$$\text{Standard Deviation: } s = \sqrt{s^2} = \sqrt{211.2} \approx 14.5$$

Consistency: No, the variability of the lifetimes of the batteries is quite large.

10. Solve the following problems for a distribution with a mean of 80 and a standard deviation of 10.
- At least what percentage of values will fall between 60 and 100?
 - At least what percentage of values will fall between 65 and 95?

Answer:

a. The interval from 60 to 100 is from $80 - 20$ to $80 + 20$, which is 20 units on either side of the mean.

Thus, $k = \frac{\text{difference}}{s} = \frac{20}{10} = 2$. Substitute into the formula: $1 - \frac{1}{k^2} = 1 - \frac{1}{2^2} = \frac{3}{4} = 0.75 = 75\%$

b. The interval from 65 to 95 is from $80 - 15$ to $80 + 15$, which is 15 units on either side of the mean.

Thus, $k = \frac{15}{10} = 1.5$. Substitute into the formula: $1 - \frac{1}{k^2} = 1 - \frac{1}{(1.5)^2} = 1 - \frac{4}{9} = \frac{5}{9} \approx 0.56 = 56\%$

11. In a distribution of 160 values with a mean of 72, at least 120 fall within the interval 67–77. Approximately what percentage of values should fall in the interval 62–82?



Answer:

Identify the given interval and proportion:

The interval 67–77 is ± 5 units from the mean (72). At least $120/160 = 75\%$ of values fall here.

Apply Chebyshev's theorem to find k and s :

Chebyshev states at least $1 - \frac{1}{k^2}$ fall within k standard deviations (s).

Set $1 - \frac{1}{k^2} \geq 0.75$, so $k^2 \geq 4$ and $k \geq 2$. The distance is 5 units, so $s = \frac{5}{k} = \frac{5}{2} = 2.5$.

Apply to the new interval: The interval 62–82 is ± 10 units from the mean. Thus, $k = \frac{10}{s} = \frac{10}{2.5} = 4$.

Calculate the percentage: At least $1 - \frac{1}{k^2} = 1 - \frac{1}{4^2} = \frac{15}{16} = 93.75\%$ fall within this interval.

